



**WOLLO University**

**Department of software engineering**

**Blueprint based exit material for**

**Software project management**

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## ❖ Understand software project management and factors that affect software quality

- **Product operation factors** – Correctness, Reliability, Efficiency, Integrity, Usability.
- **Product revision factors** – Maintainability, Flexibility, Testability.
- **Product transition factors** – Portability, Reusability, Interoperability.

Software project management is the process of planning, organizing, and managing the resources required to develop and deliver software. It involves a number of activities, such as:

- Defining the project scope: This involves identifying the goals of the project, the features that will be included in the software, and the constraints that the project must adhere to.
- Estimating the project schedule and budget: This involves estimating how long the project will take to complete and how much it will cost.
- Organizing the project team: This involves assembling a team of people with the skills and experience necessary to complete the project.
- Managing the project schedule and budget: This involves tracking the progress of the project and making adjustments as needed to ensure that the project stays on track.
- Managing the project risks: This involves identifying and mitigating the risks that could impact the success of the project.
- Communicating with stakeholders: This involves keeping stakeholders informed about the progress of the project and addressing their concerns.

The factors that affect software quality can be divided into two categories: internal factors and external factors.

- Internal factors: These are factors that are within the control of the software development team. They include the following:
  - The quality of the software requirements: The quality of the software requirements will have a direct impact on the quality of the software.
  - The quality of the software design: The quality of the software design will also have a direct impact on the quality of the software.
  - The quality of the software code: The quality of the software code will have a significant impact on the quality of the software.
  - The quality of the software testing: The quality of the software testing will help to ensure that the software is free of defects.
- External factors: These are factors that are outside the control of the software development team. They include the following:
  - The complexity of the software: The complexity of the software will have an impact on the difficulty of developing and testing the software.
  - The changing requirements: The changing requirements of the stakeholders can make it difficult to develop and deliver software that meets their needs.
  - The availability of resources: The availability of resources, such as time, money, and people, can impact the quality of the software.

By understanding the factors that affect software quality, software project managers can take steps to improve the quality of the software they develop.

Here are some additional tips for improving software quality:

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- Start with good requirements: The quality of the software requirements is the foundation of software quality. Make sure that the requirements are clear, complete, and consistent.
- Use a sound software development methodology: There are many different software development methodologies available. Choose one that is appropriate for the size and complexity of the project.
- Test the software thoroughly: Testing is essential for ensuring the quality of the software. Use a variety of testing techniques to ensure that the software is free of defects.
- Involve stakeholders throughout the development process: Stakeholders can provide valuable feedback that can help to improve the quality of the software. Make sure to involve them throughout the development process.
- Continuously improve: Software quality is an ongoing process. Once the software is released, continue to monitor and improve its quality.

## ❖ Technique and tool to support software project management

There are many techniques and tools that can be used to support software project management. Some of the most popular techniques include:

- Waterfall: This is a traditional project management methodology that breaks down a project into a series of sequential phases, such as requirements gathering, design, development, testing, and deployment.
- Agile: This is a more flexible project management methodology that breaks down a project into smaller, iterative cycles called sprints. Each sprint focuses on delivering a working product increment.
- Critical Path Method (CPM): This is a technique for scheduling and managing projects that identifies the critical path, which is the longest sequence of activities in the project.
- Work Breakdown Structure (WBS): This is a hierarchical decomposition of a project into smaller, more manageable tasks.
- Gantt chart: This is a graphical representation of a project schedule that shows the start and finish dates of tasks, as well as their dependencies.

Some of the most popular tools for software project management include:

- Asana: This is a popular project management tool that offers a variety of features, including task management, project planning, and team collaboration.
- Trello: This is a Kanban-style project management tool that allows teams to visualize their work and track progress.
- Jira: This is a popular issue tracking tool that is used by many software development teams.
- Microsoft Project: This is a powerful project management software that offers a wide range of features, including Gantt charts, resource management, and budgeting.

The best technique and tool for a particular software project will depend on the size and complexity of the project, the team's experience, and the project's specific requirements. However, the techniques and tools listed above are a good starting point for any software project manager.

Here are some additional tips for using techniques and tools to support software project management:

- Choose the right technique and tool for the project. Not all techniques and tools are created equal. Some are better suited for small projects, while others are better suited for large projects.

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- Train your team on the technique and tool. It is important that everyone on the team understands how to use the technique and tool. This will help to ensure that the project is managed effectively.
- Use the technique and tool consistently. Once you have chosen a technique and tool, it is important to use it consistently throughout the project. This will help to ensure that the project stays on track.
- Monitor and adjust the technique and tool as needed. As the project progresses, it may be necessary to adjust the technique and tool. This is especially true if the project encounters unexpected challenges.

## ❖ Software project plan and risk management plan

A software project plan is a document that outlines the goals, scope, schedule, budget, and resources for a software development project. It is a critical tool for managing the project and ensuring that it is completed successfully.

A risk management plan is a document that identifies and assesses the risks that could impact a software development project. It also includes strategies for mitigating those risks. Risk management is an essential part of software project management, as it helps to ensure that the project stays on track and that it is completed within the available resources.

The software project plan and risk management plan should be developed early in the project lifecycle, and they should be updated regularly as the project progresses. This will help to ensure that the project is managed effectively and that the risks are identified and mitigated early on.

Here are some of the key elements of a software project plan:

- Project goals: The project goals should be clearly defined and measurable.
- Project scope: The project scope should be defined in detail, and it should be clear what is included and what is not included in the project.
- Project schedule: The project schedule should be realistic and achievable.
- Project budget: The project budget should be realistic and should be based on the project scope and schedule.
- Project resources: The project resources should be identified and allocated.
- Communication plan: The communication plan should outline how information will be communicated to stakeholders throughout the project.
- Risk management plan: The risk management plan should identify and assess the risks that could impact the project.

Here are some of the key elements of a risk management plan:

- Risk identification: The first step in risk management is to identify the risks that could impact the project.
- Risk assessment: Once the risks have been identified, they should be assessed in terms of their probability of occurrence and their impact on the project.
- Risk mitigation: Strategies should be developed for mitigating the risks that have been identified.
- Risk monitoring: The risk management plan should be monitored and updated regularly to ensure that the risks are being effectively mitigated.

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By developing and implementing a software project plan and risk management plan, software project managers can increase the chances of success for their projects.

Here are some additional tips for creating a software project plan and risk management plan:

- Involve stakeholders: Stakeholders should be involved in the development of the project plan and risk management plan. This will help to ensure that the plan is aligned with the needs of the stakeholders.
- Use a project management tool: A project management tool can help to track the progress of the project and to manage changes to the plan.
- Be flexible: The plan should be flexible enough to accommodate changes that may occur during the project.
- Communicate the plan: The plan should be communicated to all stakeholders so that they understand the project's goals and objectives.

## ❖ Project management concepts to manage a software development project

There are many project management concepts that can be used to manage a software development project. Some of the most important concepts include:

- Scope: The scope of a project defines the work that needs to be done to complete the project. It is important to define the scope clearly and concisely so that everyone involved in the project understands what is included and what is not included.
- Schedule: The schedule defines the timeline for completing the project. It is important to create a realistic schedule that takes into account the complexity of the project and the availability of resources.
- Budget: The budget defines the financial resources that are available for the project. It is important to create a budget that is realistic and that will allow the project to be completed within the available resources.
- Risk: Risks are potential events that could impact the success of the project. It is important to identify and mitigate risks so that they do not prevent the project from being completed successfully.
- Communication: Communication is essential for the success of any project. It is important to communicate effectively with all stakeholders throughout the project lifecycle.
- Quality: Quality is essential for the success of any software project. It is important to build quality into the software from the start and to test the software thoroughly to ensure that it is free of defects.

These are just some of the most important project management concepts that can be used to manage a software development project. By understanding and applying these concepts, software project managers can increase the chances of success for their projects.

Here are some additional tips for managing a software development project:

- Use a project management methodology: A project management methodology is a set of principles and practices that can be used to manage projects. There are many different project management methodologies available, so choose one that is appropriate for the size and complexity of the project.
- Create a project charter: A project charter is a document that defines the scope, schedule, budget, and other important aspects of the project. It is important to create a project charter

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early in the project so that everyone involved in the project understands the project's objectives.

- Track progress and manage changes: It is important to track the progress of the project and to manage changes to the project scope, schedule, or budget. This will help to ensure that the project stays on track and that it is completed within the available resources.
- Communicate with stakeholders: It is important to communicate effectively with all stakeholders throughout the project lifecycle. This will help to keep stakeholders informed about the progress of the project and to address their concerns.
- Close out the project: Once the project is complete, it is important to close out the project. This involves finalizing the project documentation and lessons learned, and releasing the software to production.